

9. The reversible reaction shown below was one of the first to be studied in detail in the gas and liquid phases and in solution.



- (a) One study of the liquid mixture showed that it contained 0.714 % NO_2 by mass. Calculate how many moles of NO_2 would be present in 25.0 g of this liquid. Give your answer to an appropriate number of significant figures. [2]

$n(\text{NO}_2) = \dots\dots\dots$ mol

- (b) Both N_2O_4 and NO_2 are soluble in a range of solvents and the equilibrium constant, K_c , can be measured in these.

- (i) Give the expression for the equilibrium constant, K_c , for this equilibrium, giving its unit if any. [2]

$K_c =$

Unit =



- (ii) The equilibrium constants for the $\text{N}_2\text{O}_4/\text{NO}_2$ equilibrium measured at room temperature in some different solvents are listed below.

Solvent	Equilibrium constant, K_c (Unit not shown)
CS_2	17.8
CCl_4	8.05
CHCl_3	5.53
$\text{C}_2\text{H}_5\text{Br}$	4.79
C_6H_6	2.03
$\text{C}_6\text{H}_5\text{CH}_3$	1.69

- I. Samples of 0.4000 mol of N_2O_4 are dissolved separately in 1 dm³ of each of these solvents at room temperature and the reactions allowed to reach equilibrium. In one solution the concentration of N_2O_4 present at equilibrium is $5.81 \times 10^{-2} \text{ mol dm}^{-3}$. Find the value of the equilibrium constant in this solution and hence identify the solvent. [3]

Solvent

- II. The Gibbs free energy change, ΔG , of this reaction is different in different solvents. Explain how the data shows this and state, with a reason, which solvent would have the most negative ΔG value for this reaction. [2]

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